

MAT 1272 – Group Activity Answer Key

1. What are we studying?

A variable is a characteristic under investigation that assumes different values for different elements.

Classifications: major – qualitative/categorical; number of siblings – quantitative, discrete; hours studying – quantitative, continuous; favorite transportation – qualitative/categorical; height – quantitative, continuous; number of courses – quantitative, discrete.

2. Classify the variables

Variable	Q/C	D/Co
Number of students in a class	Q	D
Eye color	C	—
Amount of money in a bank account	Q	Co
Number of text messages sent yesterday	Q	D
Time needed to complete an exam	Q	Co
Type of phone owned	C	—
Weight of a student	Q	Co
Number of cars in a parking lot	Q	D

Discrete vs. continuous: Discrete variables have countable possible values; continuous variables can assume any numerical value over an interval.

3. Population and Sample

Population: All City Tech students. **Sample:** The 40 MAT 1272 students surveyed. **Variable:** Number of hours per week spent studying mathematics. **Type:** Quantitative and continuous.

4. Think like a statistician

A. Population: all City Tech students. Sample: 75 students surveyed. Variable: amount spent on lunch. Type: quantitative, continuous.

B. Population: the students the professor wants to study (e.g., students in the relevant class/population). Sample: 30 students from her class. Variable: hours spent studying. Type: quantitative, continuous.

C. Population: the students at the college. Sample: 100 students surveyed. Variable: preferred transportation method. Type: qualitative/categorical.

5. Challenge

Answers will vary. Example: favorite music genre = qualitative/categorical; number of classes taken = quantitative, discrete.